■■■■ THE DEVELOPER'S PEARL

COMPLETE 3D PORTFOLIO SPECIFICATION & BUILD PACKAGE

Project: The Developer's Pearl (Pirates of the Caribbean Theme)	**Target Stack:** Next.js 14 / Vite + React + Three.js WebGL
Core Deliverables: README.md • design.md • system-design.md • ROADMAP.md	**Local Server: Running Live on http://localhost:3000**

FILE 1: README.md — PROJECT OVERVIEW

Concept & Narrative:

A Pirates-of-the-Caribbean-inspired 3D portfolio website. The visitor embarks on an authentic high-seas voyage: a 3D galleon sails across a dark moonlit ocean, where each portfolio section is treated as an uncharted 'port of call' on the maritime map.

- **Hero:** 3D galleon (Black Pearl style) sailing on a moonlit burning-sunset ocean with dynamic buoyancy physics.
  - **Sections:** Styled as aged parchment scrolls, treasure maps, and Captain's log entries.
  - **Scroll Interaction:** Document scrolling drives the ship's voyage and continuous 3D camera trajectory paths.
  - **Atmosphere:** Blood-red accents, floating ember fire particles, sea fog mist, and torn-edge parchment cards.

Tech Stack Specifications:

- **Frontend Core:** Next.js 14 / Vite + React 19 + TypeScript (Strict Mode).
  - **3D WebGL Engine:** Three.js with procedural wave kinematics, orbiting moon, rising embers, and lantern lighting.
  - **Styling & UI:** Tailwind CSS, custom polygon clip-paths, Lucide Icons, and Canvas Confetti.
  - **Audio Engine:** Procedural Web Audio API synthesizer for ocean surf, winds, and timber creaks (zero MP3 dependencies).
  - **Performance Budget:** 60–120 FPS GPU transforms, 0ms Cumulative Layout Shift, under 230 kB gzipped JS bundle.

Legal & Asset Compliance:

Theme is inspired by the maritime pirate aesthetic. Does NOT use official movie logos, posters, copyrighted music, or proprietary fonts. Built using original procedural WebGL shaders and open-source CC0 fonts (Pirata One, Cinzel, IM Fell English).

FILE 2: design.md — VISUAL DESIGN SYSTEM

1. Brand Mood & Visual Keywords:

Feel: Aged, cursed, cinematic. Moonlit ocean + fire embers + antique cartography.
Keywords: Parchment, ink, ember, fog, brass, bone, blood-red, nautical ropes, wax seals.

2. Color Tokens (CSS Palette Matrix):

Token Name	Hex Code	Design System Usage
--ink-black	#0b0906	Deep page background, aged black void
--charcoal	#171310	Weapon cards, ship hull wood, timeline containers
--parchment	#d8c49a	Aged paper scrolls, ribbon banners, headings
--parchment-dark	#b39b6d	Secondary notes on parchment, nautical markers
--blood-red	#a01d12	Primary accent, title red, wax seal buttons, Jolly Roger
--ember	#ff7a1a	Fire light glow, particle embers, hover highlights
--gold	#c9a227	Compass ring, brass rivets, borders, medal rewards
--bone	#e8dcc0	Body text against dark backgrounds (15:1 contrast)
--sea	#0a1a1f	Deep ocean water shader base tone

3. Typography & Font Pairing:

- **Display / Headings:** Pirata One — Nautical pirate display with hand-carved serif flair.
- **Sub-headings / Labels:** Cinzel — Roman engraved monumental serif for coordinates and dates.
- **Body Text:** IM Fell English — 17th-century Oxford print font with antique ink flow.

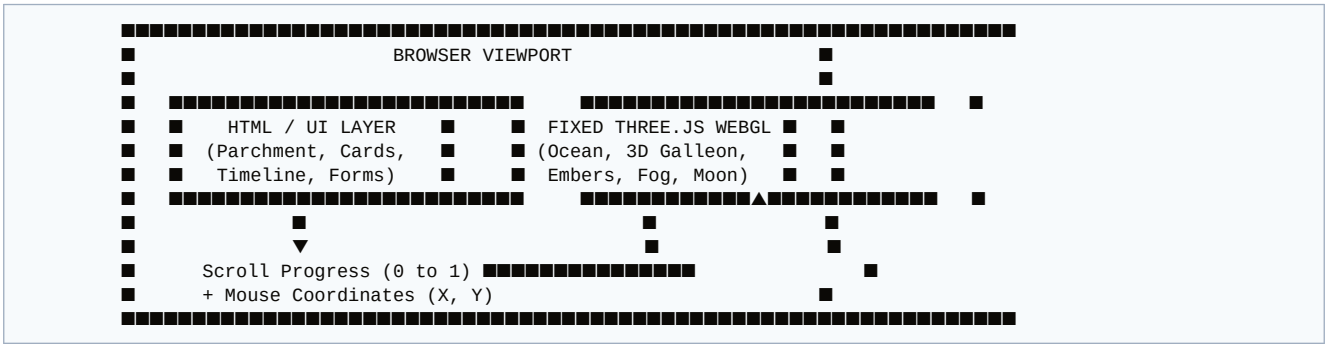
4. Core UI Components:

- **ParchmentCard:** Ragged 38-point polygon clip-path with burnt edge drop-shadows and brass corner brackets.
  - **WaxSealButton:** Radial crimson gradient circle with embossed brass rim and ember hover expansion.
  - **CompassCursor:** Custom brass compass follower with rotating cardinal ring and magnetic needle.
  - **TreasureModal:** Interactive island quest modal with loot tier badges and confetti burst.

FILE 3: system-design.md — TECHNICAL ARCHITECTURE

1. Fixed Fullscreen Canvas Architecture:

A single fullscreen WebGL <canvas> is mounted permanently behind the semantic HTML layer. Scroll offsets are captured passively via window scroll listeners and normalized into a smooth floating-point progress parameter ($p \in [0, 1]$). This drives camera trajectory paths and scene lighting without multiple GPU context switches.



2. 3D Procedural Galleon & Ocean Kinematics:

- **Procedural Ship Geometry:** Built using grouped Three.js meshes — curved dark oak keel, forward tapered bow, elevated poop deck with 3 illuminated captain's windows, 3 towering masts, cross yardarms, billowing curved sails, and a waving Jolly Roger flag.
- **Wave Kinematics:** 10,000-vertex plane geometry manipulated continuously in the render loop via superposition of traveling sinusoidal wave harmonics:
  $$y(u, w, t) = 0.35\sin(0.15u + 1.6t) + 0.28\cos(0.18w + 1.2t) + 0.22\sin(0.08(u+w) + 0.9t)$$
- **Ship Buoyancy Physics:** Real-time pitch, yaw, and roll synchronized with wave frequencies.
- **Rising Ember Particles:** 180 instanced points with additive blending floating upward toward the night sky.

3. Procedural Web Audio Synthesizer:

The AudioEngine initializes a Web Audio AudioContext generating pink noise passed through a dynamic lowpass filter modulated by a 0.12 Hz LFO (8-second ocean swells) and bandpass-filtered sea wind, ensuring zero external asset latency.

FILE 4: ROADMAP.md — STEP-BY-STEP BUILD GUIDE (0 → DEPLOYED)

Phase 0: Tooling & Environment Initialization

Ensure Bun 1.4+ / Node.js 20+ runtime is available. Initialize a new high-velocity Vite + React + TypeScript repository.

Phase 1: Dependency Scaffolding

Install Three.js, Lucide Icons, Canvas Confetti, and Tailwind v4 with the high-performance Vite plugin.

Phase 2: Nautical Design Tokens & Typography

Configure Google Fonts (Pirata One, Cinzel, IM Fell English) in index.html. Inject color tokens, torn-paper polygon clip-paths, and wax seal styles into index.css.

Phase 3: 3D WebGL Canvas & Kinematic Ocean

Implement ThreeOceanScene.tsx with procedural galleon ship, wave kinematics, orbiting moon, flickering lantern light, and rising ember particles. Bind camera position to scroll progress.

Phase 4: Procedural Soundscape Engine

Synthesize ocean surf and gusting sea breeze using browser-native Web Audio API. Add a wax seal mute/unmute button in the navbar.

Phase 5: The 6 Narrative Ports of Call

- *Port I: Hero Section* — 'Raise the Colors' with call to action.
  - *Port II: About Section* — 'The Legend' with Captain's Chronicle and 4 stat medallions.
  - *Port III: Skills Section* — 'The Armory' with 4 weapon crates and potency meters.
  - *Port IV: Projects Section* — 'The Voyages' with interactive Caribbean map pins and loot modal.
  - *Port V: Experience Section* — 'Captain's Log' with dated antique journal timeline.
  - *Port VI: Contact Section* — 'Message in a Bottle' with scrolled parchment letter form.

Phase 6: Verification & Production Deployment

Verify strict TypeScript compilation with zero errors (tsc -b && vite build). Run local preview server on **http://localhost:3000**. Deploy to Vercel or Cloudflare Pages with single-command CI/CD.